

DRIVER SAFETY MONITORING: REAL-TIME DETECTION OF DISTRACTION AND DROWSINESS

Dhanapandi V
Associate Professor

M. Kumarasamy College of Engineering
Karur, TamilNadu
dhanapandiv.cse@mkce.ac.in

Ashok E

Computer Science and Engineering
M. Kumarasamy College of Engineering
Karur, TamilNadu
ashokee359@gmail.com

Ajai R

Computer Science and Engineering
M. Kumarasamy College of Engineering
Karur, TamilNadu
ajayrathanasabapathi@gmail.com

Deepan N

Computer Science and Engineering
M. Kumarasamy College of Engineering
Karur, TamilNadu
deepannavaneetham@gmail.com

Giriprasath M

Computer Science and Engineering
M. Kumarasamy College of Engineering
Karur, TamilNadu
giriprasathlatha005@gmail.com

ABSTRACT: A significant world main contributing factor to road accident is distracted driving. Drivers who are distracted are three times more possible to get in an accident than those who are not. For this reason, enhancing road safety requires the ability to identify driver distraction. Numerous earlier researches have put forth a variety of techniques for detecting driver distraction, including approaches based on images, sensors, and machine learning. However, the accuracy, complexity, and real-time performance of these approaches are limited. This observe proposes a unique approach to motive force distraction detection with the aid of fusing the You Only Look Once (YOLO) goal identity system with a convolutional neural network (CNN). The categorization of the located objects and item detection the usage of YOLO are the 2 important levels of the proposed version. The driver's face, hands, and other items that could distract them are among the many elements in their environment that the YOLO algorithm is used to recognize and find. Next, a CNN is used to categorize the things that have been spotted in order to ascertain whether or not the driver is distracted. The suggested model performs well in identifying driver distraction when tested on a public dataset. Additionally use the CNN algorithm to analyze the driver's tiredness based on ocular attributes. The integration of the suggested technique into sophisticated driver assistance systems has promise for enhancing road safety in real-time conditions.

KEYWORDS: Yolo Algorithm, convolutional neural network, objects detection, driver distraction, and drowsiness detection.

I. INTRODUCTION

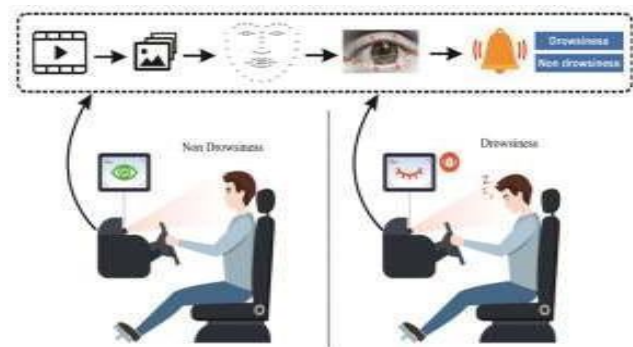
A driving force tracking system (DMS) is a part of mechanism that follows a driver's moves and uses sensors and cameras to look for symptoms of fatigue, evelopd, or impairment. Through alerting the motive force or, if required, taking corrective action, the gadget seeks to growth street protection. As a safety detail in independent cars and advanced motive force help structures (ADAS),

DMS is gaining popularity in the vehicle industry. Two of the principle reasons of road injuries are motive force fatigue and distraction. Associate to the National Highway Traffic safety Administration (NHTSA), three,142 human beings died in crashes regarding distracted using in 2019. Furthermore, up to twenty% of all visitors accidents are notion to be the result of weariness. Thus, improving road safety requires identifying and mitigating motive force weariness and distraction. A form of cameras and sensors, which include as biometric sensors, head position, face reputation, and eye monitoring, are used by DMS to identify the driving force's moves. These sensors may additionally identify symptoms of distraction, such turning far from the road or the use of a cellphone, or sign of weariness, like drooping eyelids or modifications in head function. When the motive force reveals abnormalities of their behavior, together with slurred speech or diminished motor capabilities, the system might also become aware of them and either inform them or intervene by way of directing them to pull over to the side of the road. To sum up, DMS is an crucial piece of technology that allows to improve road safety by figuring out and avoiding driving force weariness, distraction, and impairment. In the automobile quarter, the technology is spreading more widely as an ADAS and independent vehicle safety aspect.

The improvement of public protection and the lower in injuries are two of the maximum big makes use of smart shipping structures (ITS). One of the principle reasons of injuries, especially on rural roads, is fatigued riding. Fatigue impairs a driver's intelligence and ability to make choice that will preserve the automobile underneath manage. A driver's flair to manipulate the automobile, respond speedy, become aware of their environment, and perceive them is all inclined by using fatigue and sleepiness. As a result, it's miles critical that the current fashion in the automobile industry encompass motive force resource technology that

could discover driving force fatigue and sleepiness.

This take a look at provides a non-intrusive prototype computer imaginative and prescient gadget for measuring a motive force's absorption in real time. Since human eyes provide a wealth of information approximately a driver's fitness, together with gaze, note span, and level of weariness, eye song is an important device for destiny motive force assistance systems. The goal is to minimize the number of accidents and enable all drivers to travel safely. Driving at night has become dangerous, with many accidents and worries for both the general public and transportation authority's due to the rise in the movement of large vehicles. After a few days of nonstop driving, the drivers feel compelled to continue driving, which hinders their driving skills and slows down their reflexes, raising the risk of collisions. In most cases of accidents, fatigue is shown to be the main cause of sleepiness. A collection of symptoms often referred to as "fatigue" include subjective sensations of drowsiness and decreased performance. Even after a thorough investigation, the term "fatigue" still lacks a universally accepted meaning. The driver sleepiness



detection is displayed in Figure 1.

Fig 1: Drowsiness analysis

II. RELATED WORK

Ahmet Kolus, et.al,...[1] Proposed study systematically reviewed empirical studies (with reported presentation measures) on driver drowsiness detection (DDD) systems that employ eye activities to indicate drowsiness. Giving academics and professionals a comprehensive grasp of DDD systems based on eye activity was the main goal of this research. Using the suggested treatment reviews and meta-analysis methodology, 41 papers were found. This review examined several eye activity metrics for drowsiness and offers a categorization system for these metrics. Furthermore, a classification strategy for the existing ocular activity measurement methods was developed and they were analyzed. Furthermore, using the corresponding performance measures, the decision-making algorithms for classifying and predicting sleepiness states were examined. Lastly, suggestions and insights for using ocular activity measurements to identify sleepiness early on were explored. Future research and developments on DDD employing eye activities will build upon the findings of

this study. Measuring eye movement was present in about 19% of the research that this review found. Three types of measurements were identified: papillary parameters, saccadic eye movements, and eye gaze direction. The subsections provide additional information about the history, meaning, and implications of each type.

Muhammad Ramzan, et.al,...[2] Created system hybrid module's top features are investigated using the PCA control of the architecture in coordination with the proposed CDLM for detecting drowsiness. A district-wide empirical investigation was conducted using the Yawning Detection Dataset. The findings show that the suggested and implemented machine learning-based module, as well as the comparative current methodologies in the relevant literature, does not achieve the same level of accuracy as the built and designed deep learning and hybrid modules. Proposed method D3-CDLM for drowsiness detection, in this study, the proposed work consists of four modules: machine learning and ensemble methods-based, deep learning- based, and a hybrid including feature extraction and selection. The proposed method uses the benchmark dataset to classify the presence of drowsiness. Focus is on the driver's drowsiness state rather than the vehicle in behavioural measures, and the camera is used to monitor the driver's physical conditions such as eye, face, head, and yawning. Driver's eye detection, eye blink rate, yawning analysis, and facial expression analysis provide behavioural measures. This system is trained and tested by using a publicly available dataset YawDD, which is explicitly used for yawning detection using facial videos of the driver. The proposed architecture also attained 99.4% accuracy by implementing the proposed technique to these images for training and validation purposes. The hybrid approach, which combines the proposed 30 layers of CNN architecture with HOG and PCA, obtained 99.6% accuracy. The accuracy achieved by using the proposed CNN model is 99.4%, and by using machine learning, the accuracy achieved is less than the other two approaches.

Jose Alguindigue, et.Al,...[3] developed on this observe have enormous capacity for actual- global programs, mainly in commercial truck riding and fleet management structures. It produced controlled simulated riding situations with a cohort of 30 volunteers, half of whom drove in a non-monotonous and the opportunity half in an uneventful placing. Three deep analyzing techniques had been employed: in order neural network (SNN) for HRV, 1D-convolutional neural network (1D-CNN) for EDA, and convolutional recurrent neural network (CRNN) for eye monitoring. The HRV approach acquire 98.28%, 98%, and F1-score for accuracy, remember, and F1-score, respectively, at the same time as the EDA model done 96.32%, ninety six%, and ninety six% for the identical metrics. The confusion matrix affords extra perception into the model's functionality and confirms that the HRV and EDA models are effective in identify driving force drowsiness because of their excessive accurateness.

Yashar Jebraeily, et.al,...[4] The suggested strategy achieves an accuracy rate of almost 99.8%,

outperforming alternative approaches in terms of precision and accuracy. There are usually difficulties while designing and choosing a convolutional neural network structure, and these issues need to be properly handled. The number of particular layer, filter sizes, number of neurons in every layer, measure of training data, and hardware configurations are some of the significant design elements that can directly affect CNN performance. Making the wrong decisions for these factors may result in overfitting, decreased performance, or other problems. Evolutionary optimization algorithms are one of the techniques utilized in recent research to get the structure of convolutional neural networks. The main solution proposed in this explore is the use of a genetic algorithm in combination with a neural network architecture search to find an optimal structure for convolutional neural network models. This approach can design and enhance a model with high accuracy and good detection capabilities. One of the strength of this study is the utilization of a large and various dataset, including more than 7342 images of drivers during driving. This dataset allows training the network under more realistic conditions and improving model performance. Transfer learning has also been utilized to transfer valuable information obtained from the source dataset (FER-2013) to the proposed model's optimal architecture. This result demonstrates that the combined approach presented can be an effective tool for enhancing safety in driving environments and preventing road accidents. Furthermore, this research has faced challenges, such as the need for powerful hardware.

Madduri Venkateswarlu et.al,...[6] proposed a Drowsy Detect Net that utilize a shallow Convolutional Neural Network (CNN) architecture to identify driver drowsiness. To recognize faces and extract eye regions, the 68- point facial landmark identification technique is employed. With little training data, the suggested system uses a shallow CNN architecture with fewer layers and parameters to identify driver drowsiness. Eyelid closure and other pertinent visual cues for sleepiness detection are the main focus of feature extraction. Driver drowsiness is also identified using transfer learning models, including VGG19, ResNet50, MobileNetV2, and InceptionV3. Dataset-1 and Dataset-2 were the two datasets used to evaluate this research. On two datasets, the proposed Drowsy Detect Net produced an accuracy of 99.23% and 99.14%, respectively. The proposed Drowsy Detect Net framework achieved better accuracy when evaluate with state-of-the-art model and pre- skilled models. This research explored the effectiveness of utilizing a shallow Convolutional Neural Network (CNN) for driver drowsiness identification with limited training data by focusing on key visual cues like eye closure. To address the challenge of drowsiness classification, pre-trained models such as MobileNetV2, ResNet-50, InceptionV3, and VGG-19 were leveraged, and their results were compared with the proposed model. The impact of hyper parameters was also examined. It involve training the model with infrared or near-infrared images for better performance in low-light conditions and exploring diverse datasets, including night-time and adverse weather

scenarios

Shumayla Yaqoob et.al,...[7] Proposed work presents a taxonomy of schemes, and the analytical evaluation and identifies the open research challenges. Using methods like AI-based image processing, signal processing, and conventional algorithmic approaches, the work primarily looks at the modeling of driver behavior and the detection of anomalous conduct. In order to ensure safe driving, this analysis takes into account features and methodologies based on vehicle and driver health monitoring. More specifically, by examining conventional mathematical, machine learning, and deep education-based schemes, current approaches are categorized in a logical taxonomy. Additionally, a thorough analytical commentary is provided after a synopsis of the schemes is provided to emphasize the important issues. By using the information from the comparison, it helps identify research concerns. Lastly, the study emphasizes the open research difficulties that require creative answers from the researchers. Our analysis revealed two predominant data collection approaches including camera-based and sensor based. It include investigating more driving behaviors, implementing a unified AI approach for detecting multiple behaviors, creating dynamic algorithms, and minimizing sensor expenses. It shall focus on addressing driver behavior challenges and examining the correlation between behavior and environmental elements such as intersections, pedestrians, and weather. The former involves smart phone or vehicle-mounted cameras for image data; the latter involves signals from vehicles and drivers. It distinguished test environments as real scenarios for evaluating results on live data instead of relying on simulation scenarios only. We examined machine learning, deep learning methods, and traditional mathematical approaches for assessing driver behaviors.

Mehdi Sefidgar Dilmaghani et.al,...[8] suggest a method for assessing and contrasting DMS performance. The influence of pixel-bias change on the four DMS skin texture—face tracking, head location, blink counting, and gaze opinion—is examined in this work. To do this, additional measures are suggested to assess the DMS's performance both overall and for each feature. According to these measurements, stability is the most crucial component for gaze, head position, and facial tracking calculations. The primary component of this function, the blink counting accuracy, is also assessed. With the consent of the subjects, the impacts of bias variations on each aspect are investigated on several human subjects. The freshly proposed measurements are used to conclude the appropriate bias ranges for each DMS feature and the overall presentation. These feature measurements may be collective to produce another statistic that gauges the overall performance of the system. To categorise the ideal bias levels at which the DMS performs at its peak, the system's performance must be assessed using all available metrics over the whole bias assortment. This was accomplished by assigning different values to each bias throughout the whole range, and analysing each DMS property independently as well as the overall feature

combination at each value. For every value, thirty seconds of events are gathered, and each bias is assigned a range of values that fall within its range, while the default value of 0 is kept for the other biases. This procedure was carried out five times with five participants standing in front of the camera, attempting to remain still while always looking into the camera lens. These metrics offer numerical of the system's functionality, bolstered by qualitative observations.

Eunmok Yang et.al,...[9] developed process used a binary classification to reduce complexity and get general sable results by classifying two stages: alert and sleepy. It is well known that driving when significantly raises the risk of an accident. Specifically, compared to aware driving, driving while fatigued raises the risk of a collision by 4-6 times. In fact, a significant study finds that 15% to 20% of accidents involve weariness. Consequently, expanding methods that are effective in real-time monitoring of driver fatigue conditions and that can alert drivers just before they fall asleep is crucial to preventing collapses. A technique that begins monitor a driver's finding action as soon as the trip's activities commence is necessary for the detection of driver drowsiness. Signs that the driver is barely steering and short, quick, abrupt steering movements to keep the automobile on course are common markers of declining focus. Remarkably, real-time signal and image processing skills have been greatly improved with the use of deep learning (DL) approaches. The significant progress made in object identification is evidence of the evolution of deep learning methods. The automotive, medical, and security sectors are just a few of the businesses that have benefited greatly from these developments. DL application has revolutionized the identification of all types of sleepiness indicators, including biological, behavioral, and vehicle motion indicators. It is astounding how groundbreaking DL is at extraction these important indicators of drowsiness.

Eduardo Quiles-Cucarella et.al,... [10] developed approach evaluates a multidimensional sleepiness index by recording head position, direction, and facial expressions. It also investigates the feasibility of integrating it into an inexpensive package. The objective is specifically to determine the driver's state by studying their facial expressions, such as how frequently they tilt their head, blink, and yawn, open their eyes, and focus their concentration. An algorithm that can identify tiredness has been created specifically for this purpose. Facial recognition techniques based on Haar cascade and Histograms of Orientated Gradients (HOG) are compared. The Raspberry Pi, a low-cost gadget that makes it possible to build a prototype that can detect drowsiness and communicate with other camera has been used in the implementation. The findings demonstrate that algorithms based on one-index detection are not as effective in identifying fatigue as the suggested multi-index method. Biomedical signals that monitor the driver's muscles, brain, or cardiovascular indications are the foundation of drowsiness detection systems. Although non- invasive techniques have been developed to record biological

signals, electrodes in the driver are often required for signal monitoring. With tiredness, there is a noticeable increase in both the frequency and duration of closed eyelid blinks. As a result, both the blink duration and the proportion of this duration that the eyes are open decrease. If alternative model of recording bodily signals without invasive elements are discovered, this facilitates its commercial deployment.

III. EXISTING METHODOLOGIES

Numerous approaches for identifying driver distraction have been put out in the literature. These systems fall into three primary categories: machine learning-based system, image-based systems, and sensor-based systems. Image-based systems take pictures of the driver using cameras or other sensors, and then examine them to look for indications of distraction. These systems usually employ computer vision techniques to identify The driver's face, eyes, and hands and track their motions. The Face and Eyes Activity Detection (FEAD) device is an instance of an image-based machine. Another example is the Face LAB system. Using a number of cameras, it captures the driver's face and analyses their expressions to search for Warning signs of distraction. Sensor-based totally systems watch driving force behaviour and pick out indicators of attentiveness or fatigue the use of sensors such as electroencephalography (EEG), gyroscopes, and accelerometers. For instance, Volvo's weariness the detection system tracks the motive force's grip energy and appears for indicators of fatigue the usage of sensors inside the steerage wheel. False positives, in which the gadget detects distraction even if the driver isn't always, are one of the number one issues. A variety of things like as variations in the Lighting, different people in the vehicle, or the movements of the driver, would possibly bring about fake positives. Another undertaking is real-time overall performance, especially for independent motors where the machine have to be capable of react quickly to capacity dangers. This requires excessive accuracy and occasional latency, however obtaining those can be difficult, particularly in complicated using eventualities. The modern basis for hardware-based totally implementation is depicted in Figure 2.

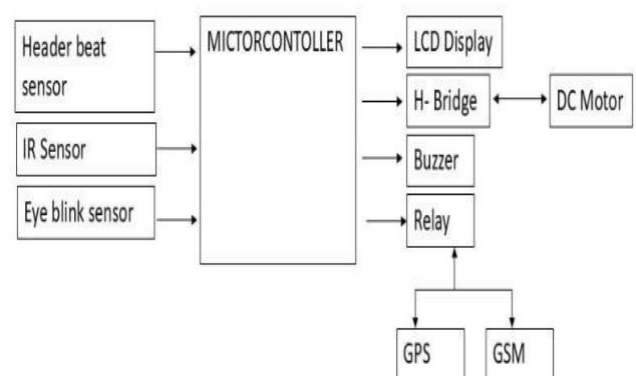


Figure 2: Existing Framework

A system called driver disturbance detection watches a driver's actions and look for indicator of distraction using a variety of sensors and algorithms. Whether a motorist is distracted by an internal issue like

sleepiness or an external one like using a phone, the system can tell when they are not paying enough attention to the road.

IV. PROPOSED METHODOLOGIES

A system called driver disturbance detection watches a driver's actions and look for indicator of distraction using a variety of sensors and algorithms. Whether a motorist is distracted by an internal issue like sleepiness or an external one like using a phone, the system can tell when they are not paying enough attention to the road. There are several advantages when comparing the proposed method to the driver monitoring systems that are in place. First off, it uses a arrangement of object detection techniques and machine learning to accurately and consistently identify signs of fatigue and distraction. Second, the system is designed to operate in immediate with low latency, ensuring that possible threats are speedily identified and dealt with. Finally, because the gadget is easy to integrate into current automobiles, it is a cost-effective

solution to improve road safety. However, there are also a number of limitations and issues with the proposed approach. One of the main challenges is that the YOLO algorithm needs high-quality camera pictures to recognize the driver's face and eyes accurately. Therefore, making sure that the dataset used to educate the CNN set of rules correctly replicates actual-world using situations is important. There is a chance that the proposed YOLO and CNN set of rules-based totally technique for perceive motive force fatigue and interest could improve road safety. However, extra research and development are had to address the constraints and demanding situations associated with the machine and make certain its effectiveness in actual- international using eventualities. The optionally available machine architecture for motive force distraction and drowsiness detection is proven in Fig. 3.

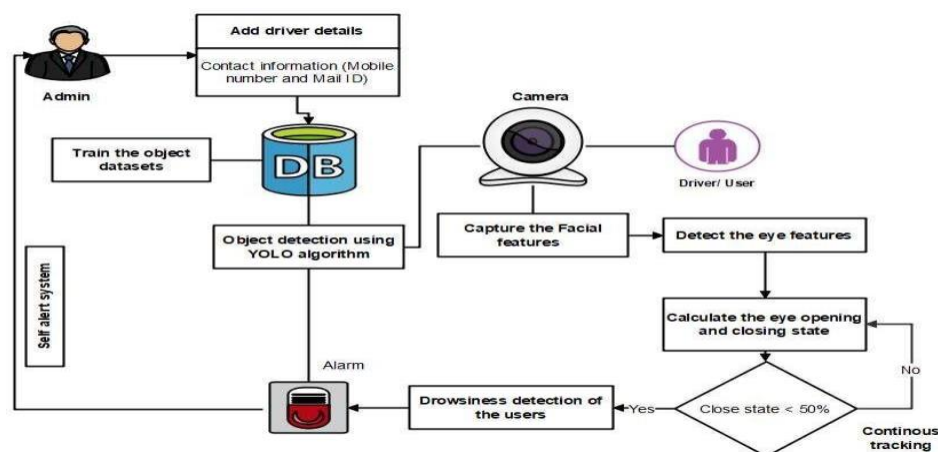


Figure3: System Architecture

The ratio of eye width to eye length is known as the eye issue ratio (EAR), and it is able to be used to gauge what number of the eyes open and near.

EAR values adjust with the scale of the attention; when the eye is display, EAR value is more advantageous, and while the eye is closed, EAR price tends towards zero.

Firstly, the attention curve is placed, and then the EAR price is designed via the attention key point, and the attention fame of the detect person is judged by using the alternate in EAR cost.

The positioning of the left eye key factor are $p_{60}-p_{67}$

The formulation for EAR computation is as follows:

$$EAR = \frac{\|P_{61}-P_{67}\| + \|P_{63}-P_{65}\|}{2\|P_{60}-P_{64}\|} \quad (1)$$

After extra variety of exams and data checks, it was found

that detection accurateness and balance were better while the EAR charge was set to 0.15, so the threshold was set to 0.15. When EAR is much less than zero.15, it could be choose that the eye's state is closed.

YOLO ALGORITHM STEPS

Step 1: YOLO divide an input parent into an $S \times S$ grid.

Step 2: If the midpoint of an entity falls into a grid mobile, that grid cellular is responsible for detecting that object.

Step 3: Each grid cell predicts B sure boxes and self-guarantee rankings for those packing containers. These poise rankings reproduce how satisfied the version is that the container comprise an item and the way correct it think the are expecting container is.

Step 4: YOLO are expecting a couple of bounding bins in line with grid cellular. At education time, we handiest want one bound box forecaster to be liable for each object. YOLO

assigns one detects to be “accountable” for anticipate an object based on which prediction has the primary current IOU with the floor truth.

Step 5: Each prophet receives better at forecasting positive sizes, characteristic ratios, or instructions of remember, humanizing the overall recollect rating.

V. EXPERIMENTAL RESULTS

In this segment used actual time datasets. This scaffold used the drowsiness detection technique. Then can appraise the concert using accurateness metrics. The accurateness metric is evaluated as

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} * 100 \quad (2)$$

The proposed algorithm offer better accuracy price than the device learning algorithms.

Accuracy table proven in desk 1.

Algorithm	Accuracy(%)
Navive bayes	20
Support Vector Machine	50
Convolutional Neural Network	70

Table (1) Accuracy desk

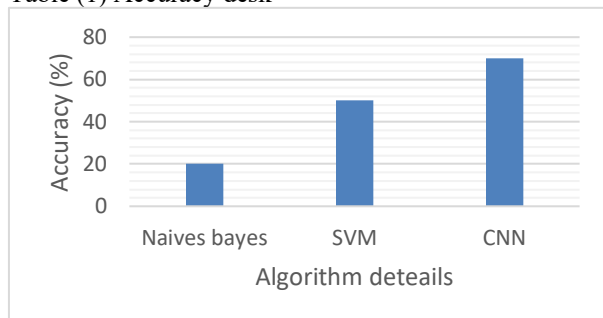


Fig 4: Performance record

From the performance chart, CNN offer improved degree accurateness than the existing gadget mastering algorithms. The proposed gadget shown lessen wide variety fake tremendous price.

VI. CONCLUSION

To sum up, research on driver distraction detection is essential since it has the potential to avert collisions and save lives while driving. The suggested method for detecting driving distraction and tiredness utilizing CNN and YOLO algorithms may be a useful way to spot possible risks and instantly notify drivers of them. The YOLO model is educated the use of the COCO dataset, and the camera is utilised to capture pix of the driver and analyse their behaviour. While the YOLO version is used to stumble on gadgets, the CNN set of rules is used to examine the driver's behaviour and locate signs and symptoms of fatigue and attention. Additionally, enhancing the accuracy of the object identification set of rules can benefit the proposed

gadget. This may be executed with the aid of hire a pre-skilled shape that has been trained on a larger dataset, or by using increasing the dimensions of the schooling dataset. Combining the system with different state-of- the-artwork technologies, which includes synthetic intelligence, gadget gaining knowledge of, and pc vision, can also enhance it. Future research on motive force attention detection may also recognition on incorporating strategies apart from visual ones. To locate auditory interruption, as an instance, system studying algorithms may examine the noises in the automobile or the motive force's speech style. Movement-detecting sensors, such as gyroscopes or accelerometers, may be used to identify tactile interruption, such the use of a telephone while using.

REFERENCES

- [1] Kolus, Ahmet. "A systematic review on driver drowsiness detection using eye activity measures." IEEE Access (2024).
- [2] Ramzan, Muhammad, et al. "A Novel Hybrid Approach for Driver Drowsiness Detection using a Custom Deep Learning Model." IEEE Access (2024).
- [3] Alguindigue, Jose, et al. "Biosignals Monitoring for Driver Drowsiness Detection using Deep Neural Networks." IEEE Access (2024).
- [4] Jebraeily, Yashar, Yousef Sharafi, and Mohammad Teshnehlab. "Driver Drowsiness Detection Based on Convolutional Neural Network Architecture Optimization Using Genetic Algorithm." IEEE Access (2024).
- [5] Madni, Hamza Ahmad, et al. "Novel Transfer Learning Approach for Driver Drowsiness Detection Using Eye Movement Behavior." IEEE Access (2024).
- [6] Venkateswarlu, Madduri, and Venkata Rami Reddy Ch. "DrowsyDetectNet: Driver Drowsiness Detection Using Lightweight CNN with Limited Training Data." IEEE Access (2024).
- [7] Yaqoob, Shumayla, et al. "AI-Driven Driver Behavior Assessment through Vehicle and Health Monitoring for Safe Driving-A Survey." IEEE Access (2024).
- [8] Dilmaghani, Mehdi Sefidgar, et al. "Optimization of Event Camera Bias Settings for a Neuromorphic Driver Monitoring System." IEEE Access (2024).
- [9] Yang, Eunmok, and Okyeon Yi. "Enhancing Road Safety: Deep Learning-Based Intelligent Driver Drowsiness Detection for Advanced Driver-Assistance Systems." Electronics 13.4 (2024): 708.
- [10] Quiles-Cucarella, Eduardo, et al. "Multi-Index Driver Drowsiness Detection Method Based on Driver's Facial Recognition Using Haar Features and Histograms of Oriented Gradients." Sensors 24.17 (2024): 5683.
- [11] Deng, Wanghua, and Ruoxue Wu. "Real-time driver-drowsiness detection system using facial features." IEEE Access 7 (2019): 118727-118738.
- [12] Guede-Fernandez, Federico, et al. "Driver drowsiness detection based on respiratory signal analysis." IEEE access 7 (2019): 81826-81838.
- [13] M Aarthi, R Mahima, S Sujanthi, AM Santosh, Sithara Sekar, JR Matthew, C Rooban, Ramamoorthy Ramamoorthy "Driver drowsiness detection using deep learning technique" pp.020248-1 to pp.020248-9
- [14] Santhiya S; Divyabharathi T; Jeno J; Kawin D, "Improved Authentication and Drowsiness Detection from Facial Features using Deep Learning Framework in Real Time Environments," 2023 9th International Conference on Advanced Computing and Communication Systems (ICACCS), Coimbatore, India, 2023, pp. 1638-1642, doi: 10.1109/ICACCS57279.2023.10112846
- [15] V. Karthikeyan, E. Raja & D. Pradeep," Energy based denoising convolutional neural network for image enhancement" , The Imaging Science Journal,2023, DOI: 10.1080/13682199.2023.2198350